

Xujiang Tang

Undergraduate Researcher in Mathematics and Machine Learning

txj_262538@163.com | txj2006.github.io | github.com/TXJ2006 | ORCID: 0009-0008-1127-6420

RESEARCH PROFILE

Undergraduate researcher working across mathematically grounded machine learning, optimization, nonlinear dynamics, and stochastic modeling. Current interests include bilevel optimization, online learning and bandit algorithms, Koopman operator methods, rough volatility, and sequential decision-making.

EDUCATION

B.Sc. in Mathematics and Applied Mathematics

Sep 2023 - Jun 2027

Yangtze University College of Arts and Sciences, Jingzhou, China | GPA: 3.5/4.0

- Optimization Theory: 97/100, ranked first in the major; Numerical Computation Methods: 96/100.
- Coursework: mathematical analysis, advanced algebra, probability and mathematical statistics, differential equations, number theory, mathematical modeling, and mathematical software.

Visiting Student

2025 - Present

The Hong Kong University of Science and Technology (Guangzhou) | Advisor: Prof. Tianyuan Jin

- Theoretical machine learning, online learning, bandit algorithms, stochastic optimization, and mathematical research.

PUBLISHED AND ACCEPTED

1. Q. Li and **X. Tang**. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. *IEEE ICMEIM 2025*, pp. 1-5. EI-indexed.
2. **X. Tang** and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. *IEIT 2025*. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
3. Q. Li and **X. Tang**. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. *AIMS Mathematics*. Accepted.

UNDER REVIEW AND SUBMITTED

1. **X. Tang et al.** Spectral Network Determinants of Seizure-like Synchronization and Spread in Coupled Hindmarsh-Rose Brain Models. *Cognitive Neurodynamics*. Under review.
2. **X. Tang et al.** Structure-preserving Koopman Predictive Control for Memristive Neural Dynamics: Input-exact and Commutator-defect Lifting. *Biological Cybernetics*. Under review.
3. Y. Fan, **X. Tang**, and R. Guan. Deep Hedging under Rough Volatility: A Fractional Kernel Embedding Approach with Optimal Convergence Rate. *AIMS Mathematics*. Under review.
4. **X. Tang et al.** The Curvature Filter in Bilevel Optimization: Implicit Differentiation Without Nondegeneracy. *Transactions on Machine Learning Research*. Submitted.

RESEARCH EXPERIENCE

Bilevel Optimization and Implicit Differentiation <i>Independent Research</i> - Investigated hypergradient computation beyond invertible-Hessian assumptions using differentiable lower-level solution manifolds. - Developed a curvature-filter framework based on projection operators and singular behavior near focal sets.	2025 - Present
Deep Hedging under Rough Volatility <i>Independent Research</i> - Developed a fractional-kernel neural architecture for variance-optimal hedging under rough-volatility models. - Integrated stochastic analysis, neural approximation, and numerical optimization.	2025 - Present
Dynamical Systems and Koopman Operator Learning <i>Research Collaborator, Prof. Yanbing Jia's Group</i> - Proposed a Koopman Operator Linearization Skeleton for long-horizon nonlinear prediction. - Implemented and evaluated the framework on multiple chaotic-system benchmarks against deep-learning baselines.	Mar 2024 - Present
Biomedical Data Science (Spatial Transcriptomics) <i>Research Assistant, Northeastern University (Shenyang) Supervisor: Dr. Dayu Hu</i> - Developed clustering methods and preprocessing workflows for high-dimensional single-cell data. - Built data-processing pipelines for downstream biological analysis.	Sep 2024 - Jan 2025
Peking University Open-Source LLM Group <i>Research and Engineering Intern</i> - Contributed to large-language-model deployment and retrieval-augmented generation systems. - Implemented retrieval and knowledge-integration modules for external knowledge use.	Jun 2024 - Aug 2024
Stochastic Modeling and Quantitative Finance <i>Team Lead, Student Research Project</i> - Led work on rough-volatility models, Bayesian inference, stochastic simulation, and MCMC parameter estimation.	Oct 2023 - Apr 2024

HONORS

Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge | Regional Gold Medal, WorldQuant International Quant Championship | Third Prize, UCAS Graduate AI Forum | National Second Prize, National College Students Statistical Modeling Competition | National Second Prize, Hua Zhong Cup Mathematical Modeling Competition

SKILLS

Programming and tools: Python, PyTorch, NumPy, Pandas, C++, R, MATLAB, Git/GitHub, Linux, LaTeX.
Methods: machine learning, statistical modeling, optimization, causal inference, LLM deployment, retrieval-augmented generation.
Languages: Mandarin Chinese; English.